

## Markscheme

Mathematics: analysis and approaches  
Practice question bank

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147. (a)

Points  $z_1 = 2 + i$  and  $z_2 = 5 + 5i$  represent two vertices of a square on an Argand diagram, with  $z_1z_2$  as one side.

Find the exact length of  $z_1z_2$ , and hence find the area of the square.

$$|z_2 - z_1| = |3 + 4i| \quad (M1)$$

$$= 5 \quad \mathbf{A1}$$

$$\text{area} = 5^2 \quad (M1)$$

$$= 25 \quad \mathbf{A1}$$

**[5 marks]**